

Josephus Problem II

Time limit: 1.00 s

Memory limit: 512 MB

Consider a game where there are n children (numbered $1, 2, \dots, n$) in a circle. During the game, repeatedly k children are skipped and one child is removed from the circle. In which order will the children be removed?

Input

The only input line has two integers n and k .

Output

Print n integers: the removal order.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq k \leq 10^9$

Example

Input	Output
7 2	3 6 2 7 5 1 4